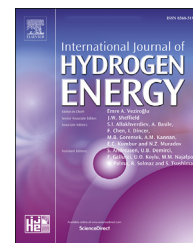


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Experimental investigation of synthesis gas production by methane reforming with flue gas over a NiO-Al₂O₃ catalyst: Reforming characteristics and pressure drop

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ABSTRACT

An experimental investigation of methane reforming with flue gases obtained by combustion of methane with the ratio of air to methane of 1 is carried out over NiO-Al₂O₃ catalyst. The set of experiments was conducted to understand the effects of operational parameters such as temperature, pressure, residence time and inlet gas composition on the conversion of the reactants and the yields of products. It is found that at the residence time above 4 kg_{cat} s/mol the methane conversion has a value close to equilibrium. A change in pressure from 1 to 5 bar has a negligible effect on methane conversion, not more than 1–2% for various operating parameters. The steam addition to the inlet reaction gas mixture increases the methane conversion considerably; especially at the temperature range from 500 to 800 °C. The steam addition to the inlet mixture has a notable on synthesis gas composition, especially at the temperature range below 700 °C. Moreover, the experiments have been carried out to determine the pressure drop and the loss factor in a packed-fixed bed of a methane reformer. A characteristic quadratic dependence of the pressure loss versus velocity $\Delta P = f(v^2)$ is observed for all experiments.

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Introduction

Synthesis gas is a fuel gas mixture consisting mainly of hydrogen and carbon monoxide. Synthesis gas (or syngas) is widespread in various industries: energy, chemical processes, transport, etc. The main process for syngas production is reforming of hydrocarbons with various oxidizing agents, such as, steam, carbon dioxide, etc. It is known that steam reforming of natural gas is the main process for the syngas production [1–6]. Methane reforming with carbon dioxide is widely used to produce synthesis gas with a high CO/H₂ ratio

[7–10]. But steam methane reforming is more widely used process for synthesis gas production [1]. As is known, the majority of hydrogen ($\approx 90\%$) is produced from natural gas by steam reforming [1]. Also, combined steam and dry reforming of methane (the main component of natural gas (up to 98–99%)) is used in the chemical process industry to produce synthesis gas with a given composition [11–14]. Moreover, gasification of coal and biomass is widely used to produce synthesis gas with high carbon monoxide content [15,16]. The fossil fuel reforming processes are usually accompanied by water-gas shift reaction which can be considered as a separate way of synthesis gas production [17–19].

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In recent times, researchers and scientists from different countries have paid considerable attention to the use of methane reforming to improve the energy efficiency of fuel-consuming plants [20–27]. Such a method of increasing energy efficiency is called thermochemical recuperation. The main concept of thermochemical recuperation (TCR) is the transformation of exhaust gases heat into chemical energy of a new synthetic fuel that has higher calorimetric properties such as low-heating value [28,29]. In fuel-consuming plants with thermochemical waste-recuperation, synthesis gas is used as the main fuel.

In the literature have been published a large number of studies devoted to synthesis gas production by various reactions. In general, scientific publications are devoted to the design and performance study of the new catalysts for improving efficiency of syngas production [30–34], to the energy and exergic analysis of the reforming processes for the different schemes [2,35,36], to the numerical modeling of methane reforming processes [33,37–40].

Among the main disadvantages of steam and carbon dioxide reforming of methane for syngas production can be identified the need for the production of oxidants and their heating to high temperatures [22,41,42]. For example, steam production for steam reforming of methane requires more than 10% of the heat spent for the process [22,41]. One solution to this problem is the use of flue gases as methane oxidants, which contain steam and carbon dioxide. This method of methane reforming has attracted interest from both industrial and environmental perspectives. Firstly, from an environmental point view, the two most abundant carbon containing greenhouse gases, methane and carbon dioxide, can be utilized effectively in this reaction and converted into useful chemical products. Secondly, at present, this methane reforming method has found wide application for the thermochemical heat recovery system.

Combined steam-dry methane reforming was studied both theoretically and experimentally. Ozkara-Aydinoglu conducted thermodynamic analysis of combined steam-dry reforming of methane to synthesis gas at the wide range of operational parameters such as temperature, pressure and inlet ratio $\text{CH}_4/\text{H}_2\text{O}/\text{CO}_2$ [11]. The author has established that the maximum methane conversion is achieved at a temperature of 900–1000 K for different initial compositions and pressures.

Al-Nakoua and El-Naas reported results of a numerical investigation of combined steam-dry methane reforming in a fluidized bed membrane reactor [43]. They described the influence of key parameters on reactor performance including reactor temperature, reactor pressure, steam to methane feed ratio (S/C) and carbon dioxide to methane feed ratio (CO_2/C).

Lemonidou and Vasalos experimentally investigated combined CO_2 methane reforming with steam methane reforming over 5 wt.% $\text{Ni}/\text{CaO}-\text{Al}_2\text{O}_3$ catalyst [44]. They investigated steam addition to the carbon dioxide methane reforming process to test the increase of the conversion of methane and the decrease of carbon deposition in catalyst. Industrially, both steam and dry methane reforming are mostly performed over nickel supported Al_2O_3 catalyst around 1000–1100 K. However, the catalyst is mostly deactivated by carbon deposition onto its surface, which is the major problem to catalyst stability. Khania and co-authors presented

article with results of experimental investigation of ZnLaAlO_4 supported Ni, Pt and Ru nanocatalysts applied for combined dry-steam reforming of methane [45].

In most of the works mentioned above, the focus is on investigation of the reforming characteristics while the important task of determining pressure losses remains unaffected. For engineering calculations (especially the thermochemical waste-heat recuperation systems [46]), in addition to the characteristics of reforming, it is necessary to know the pressure loss in the catalyst bed. This results will help to determine the pressure drop in the methane reformers as well as outlet gas composition and reaction enthalpy of the methane reforming process.

At the same time, the available publications do not cover all the practically interesting cases dealing with the reforming of methane by the products of its complete combustion (flue gases). Most of the works devoted to the experimental study of reforming used pure steam and carbon dioxide in the presence of nitrogen as oxidants. For example, to conduct engineering calculations of the thermochemical waste-heat recuperation systems, it is necessary to have accurate data on methane reforming with flue gases in order to design and optimize thermochemical reactors, as well as perform an energy analysis of the TCR systems.

In the presented article, an experimental study of syngas production by methane reforming with flue gases (obtained by the complete combustion of methane) was carried out for various operating parameters such as temperature, pressure and $\text{CH}_2/\text{CO}_2/\text{H}_2\text{O}$ inlet composition, as well as the residence time. Kinetic data were obtained, which made it possible to establish the dependence of the methane conversion on residence time for different temperatures, pressures and inlet gas compositions. The pressure drop and the loss factor in a packed-fixed bed of a methane reformer was determined for the wide range of operational parameters. In this study, flue gases as the methane oxidant were obtained by methane combustion at an excess air ratio of 1.0.

Experiments

Experimental setup

A schematic diagram of the experimental setup is presented in Fig. 1. The experimental setup can be divided into three blocks. The first, the block for preparing of the reaction mixture. The second, the block of the reforming process. The third, the block for measuring and storing of the results. The main elements of the first block are an electrical steam generator and a gas burner. The main element of the second block is a methane reformer. The main component of the third block is a portable gas analyser.

The experimental setup mainly consists of the following elements: 1 - a methane reformer; 2 - an electrical muffle furnace; 3 - a watt-hour meter; 4 - the thermocouples; 5 - a portable gas analyser; 6 - a personal computer; 7 - the mass-flow meters; 8 - a tangential burner; 9 - an electrical steam generator; 10 - a gas intercooler; 11 - a manometer.

The methane reformer made of high-alloy steel has been placed in an electrical muffle furnace. The electrical furnace

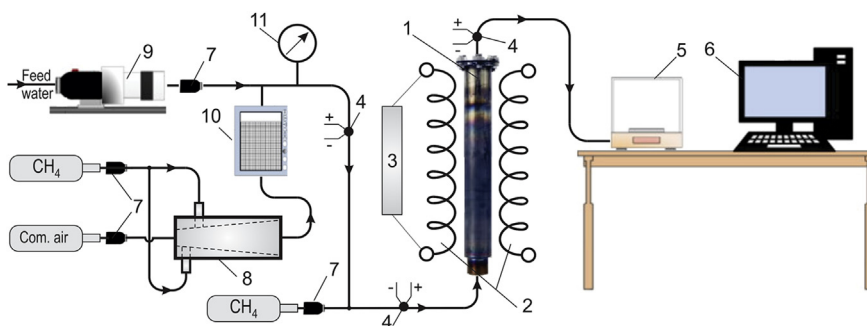


Fig. 1 – Schematic view of experimental setup.

heats the reaction space up to 1000°. The thermocouples are made of a tungsten-rhenium alloy. The gas composition is controlled by the portable infra-red as analyser Gasboard 3100P, which can be connected to a personal computer. In a leakproof gas burner with a tangential admission of methane, methane combustion takes place with an excess air factor equal to unit. Superheated steam is produced in the electric steam generator PG-LEK 1.1 (in Russian). High purity methane is used as a fuel and as a reagent for reaction reforming in the experiments, as well as a high purity compressed air is used for combustion. High purity distilled water is used for a steam production.

A photograph of the experimental setup and some elements of the schematic diagram are shown in Fig. 2. The numbers of the elements in Fig. 2 coincide with the figure caption to Fig. 1.

Catalyst

The reaction space of the methane reformer is filled with Ni-Al₂O₃ catalyst. For experiment the Ni-Al₂O₃ catalyst is used with the following chemical composition: NiO = 14.5%; SiO₂ = 0.2%; support CaO-MgO-La₂O₃-Al₂O₃ [47]. Other properties are presented below: bulk density is 680 kg/m³; surface

area is 4.5 m²/g_{cat}; average porosity of catalyst particles is about 41%. Cylinder with 7 holes is used a catalyst for experiment.

The catalyst particles are placed into the tube with diameter of 100 mm. The catalysts layer depth is 600 mm. The geometrical characteristics of catalysts are shown in Table 1.

SEM image of catalyst is presented in Fig. 3 (right side). Scanning Electron Microscopy (SEM) photo of catalyst structure is provided by the catalyst manufacturer NIAP-CATALYZATOR LLC (Novomoskovsk, Russia). The author thanks the manufacturer NIAP-CATALYZATOR LLC for the SEM image.

Experimental procedure

In the present study, a set of experiments was performed for various operational parameters: a temperature range from 400 to 800°C, a pressure from 1 to 5 bar, a composition of the inlet reaction mixture, a residence time. The residence time was varied due to a change in the flow rate of the reaction mixture. A set of experiments was performed when steam is added to the flue gases.

Before the experiment, the electric furnace heats the catalyst to the predetermined temperature. When the catalyst is heated, the reaction space of the methane reformer is filled with nitrogen. Heating of the catalyst takes approximately 15–17 min. During this period, the temperature in the reformer stabilizes throughout the reaction space. In parallel with the heating of the catalyst, combustion of methane is regulated by means of flowmeters. The temperature of the combustion products is regulated in the heat exchanger to a temperature that is higher than the catalyst temperature in the reformer. This is due to the fact that when the methane for reforming reaction is fed into the combustion products, the temperature of the mixture will decrease. The target temperature of the flue gases after the burner is determined experimentally in such a way that the temperature of the reaction mixture before the reformer is equal to the temperature of the reaction space of the reformer.

At the initial time, the system of flue gas supply to the reaction space is not connected to the reformer. After the reformer becomes warmed up and the combustion of methane is regulated, the flue gas supply system is connected to the reformer. With the help of an electric heater, the temperature in the reaction space during the experiments is regulated. Each experiment has a duration of 10–15 min.

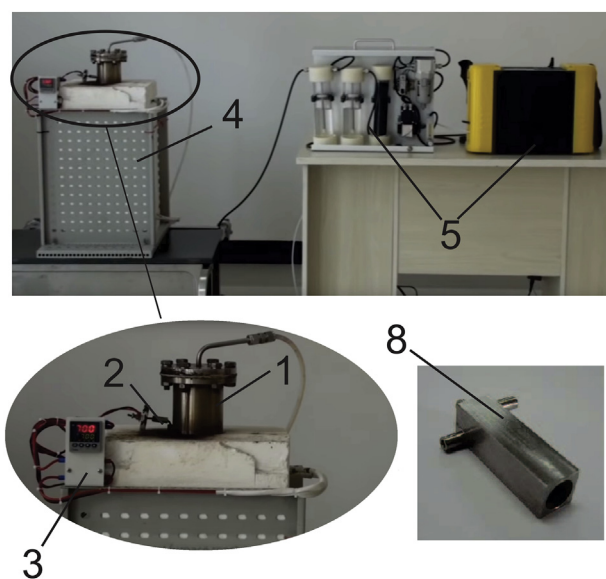
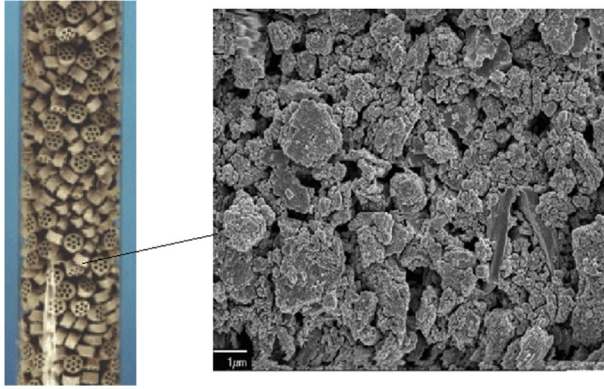


Fig. 2 – Photograph of experimental setup.

Table 1 – The geometrical characteristics of catalyst.

Catalyst shape	Diameter, mm	Thickness, mm	Holes, number/diam.	Geom.surface, m ² /m ³	Interpartical porosity m ³ /m ³
Cyl. with holes	25	25	7 × 3	347	0.46–0.49

**Fig. 3 – Left side are photographs of the packed beds filled by the catalysts with cylinders with 7 holes. Right side is SEM image for Ni-Al₂O₃ catalyst.**

During the experiment, the composition of the reformed products and the heat consumption for the reforming reaction are fixed every minute.

The fresh catalysts are used for each experiment.

Results and discussion

To analyse the effectiveness of the methane reforming process, the conversions of reactants and the yield of main products of reformat, both hydrogen and carbon monoxide, are defined as follows:

– methane conversion:

$$X_{CH_4} = \frac{CH_{4(in)} - CH_{4(out)}}{CH_{4(in)}} \quad (1)$$

– carbon dioxide conversion:

$$X_{CO_2} = \frac{CO_{2(in)} - CO_{2(out)}}{CO_{2(in)}} \quad (2)$$

– steam conversion:

$$X_{H_2O} = \frac{H_2O_{(in)} - H_2O_{(out)}}{H_2O_{(in)}} \quad (3)$$

– hydrogen yield:

$$Y_{H_2} = \frac{2 \cdot H_2(out)}{4 \cdot CH_{4(in)} + 2 \cdot H_{2(in)}} \quad (4)$$

– carbon monoxide yield:

$$Y_{CO} = \frac{CO_{(out)}}{CO_{2(in)} + CH_{4(in)}} \quad (5)$$

For further analysis, the parameter β is used. Here, β is the flue gases obtained by combustion of 1 mol of methane at excess air factor of 1.0. With complete combustion of 1 mol of methane, beta is equal to: $\beta = 2H_2O + CO_2 + 7.52N_2$.

Methane conversion

The maximum methane conversion can be achieved at equilibrium conditions. The equilibrium conversion of methane depends on the temperature, pressure and composition of the inlet reaction mixture, and is independent of the type of catalyst, residence time and other process parameters. To estimate conversion of methane obtained in the experiments, the equilibrium conversion of methane is calculated by using Gibbs free energy minimization method. For thermodynamic analysis a commercial software ASPEN-HYSYS is used. The algorithm of calculation the equilibrium gas composition is presented in the author's previous works [22,29]. For detailed analysis of the methane reforming process, parameter “residence time” is introduced. Weight residence time W_{cat}/F_{CH_4} is calculated as catalyst pellet weight divided by the feed rate of the methane and expressed as $kg_{cat}s/mol$. The residence time shows a mass of catalyst bed per mole flow-rate on methane inlet.

Fig. 4 shows changes of the methane conversions as a function of temperature when various composition of inlet reaction mixture is fed into the methane reformer at pressure of 1 bar for the different residence times. Hollow points show the experimental results; solid lines are the corresponding fitting curves. Two compositions of the reaction mixture were investigated $CH_4:1/3\beta = CH_4:0.67H_2O:0.33CO_2:2.51N_2$ and $CH_4:2/3\beta = CH_4:1.34H_2O:0.66CO_2:5.02N_2$. For each case, the residence time was chosen as close as possible to each other, however, due to of the increase of the mass flow rate of the reaction mixture through the reformer, minor discrepancies were observed.

Fig. 4 shows the positive effect of temperature on CH_4 conversion. As the temperature increases, the methane conversion increases for the each residence time. When the residence time is low, the methane conversion is almost proportional to temperature. However, the effect of temperature on CH_4 conversion has non-linear relationship at the high residence time. Moreover, the addition of extra flue gas to the feed stream increased the methane conversion considerably by 6–11% for the different temperatures. For a high residence time, the methane conversion tends to an equilibrium value.

Fig. 5 depicts the methane conversion as a function of the residence time at different temperature, $CH_4:1/3\beta$, pressure 1 bar. Dashed lines denote the equilibrium methane conversion for each temperature. It is seen that at the fixed temperatures, the methane conversion increase with increasing

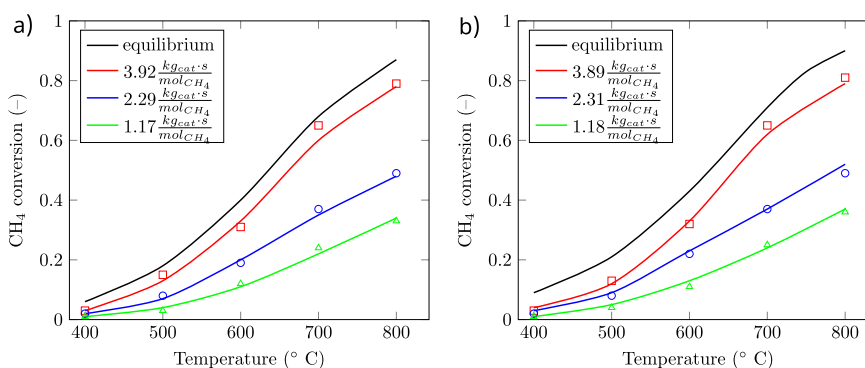


Fig. 4 – Effect of temperature on the methane conversions at $p = 1$ bar for the various inlet gas compositions: a) $\text{CH}_4:1/3\beta$; b) $\text{CH}_4:2/3\beta$. Hollow points – experimental results; solid lines – approximation curves.

residence time. However, the curves also show that as the residence time increases to higher than $4 \text{ kg}_{\text{cat}} \text{ s/mol}$ in the temperature range higher than 700°C , the methane conversion increases slightly. Obviously, with a further increase in the residence time, the methane conversion reaches a maximum and stable value because the methane reforming process reach to close to equilibrium condition.

Table 2 shows the changes in the steam and carbon dioxide conversions for the methane reforming process at a fixed $\text{CH}_4:1/3\beta$ ratio. It is seen that keeping the methane-to-oxidant ratio at 1.0 the reactant conversion tends to unity with increasing temperature. However, in practice, the use of methane-to-oxidant ratio equal to 1.0 is difficult, because even with an experiment duration of about 15 min, traces of soot carbon are observed on the surface of the catalyst after experiments. The carbon formation in methane reforming by flue gases is the subject of a separate author's study.

H_2 and CO yields

The main components of the products of the methane reforming process are hydrogen and carbon monoxide. The H_2

and CO yields for the methane reforming process are presented in Fig. 6. For the all inlet gas compositions, increasing temperature led to an increase in the hydrogen yield. However, upon addition of the flue gas to the feed stock, the trends changed. If the hydrogen yield monotone increase with an increasing of temperature for the all residence time range, than for the high temperature and the high residence time the hydrogen yield reaches a maximum value and then decreases with increasing temperature. A maximum of H_2 yield for $\text{CH}_4:2/3\beta$ is reached around temperature of $700\text{--}800^\circ\text{C}$ for the high residence time. Decrease in hydrogen yield at high temperature may be two reasons. Firstly, at high temperatures dry methane reforming reaction, which yields less hydrogen compared to steam methane reforming, and water-gas shift reaction, which simultaneously consumes hydrogen, are favoured (because as is known that the system is thermodynamically favorable when its total Gibbs free energy is at its minimum value. For a high temperature, the change in Gibbs free energy for DMR reaction is less than for SMR reaction). Secondly, the addition flue gases over the stoichiometric amount of methane, steam and carbon dioxide in the inlet reaction mixture is shown in the denominator of the hydrogen yield equation Eq. (4), which also lowered the resulting hydrogen yield values. In conclusion, all these results indicate that proper control of the residence time and the inlet gas mixture and temperature is essential to obtain the maximum hydrogen yield.

Fig. 7 shows the dependence of the carbon monoxide yield on temperature. With increasing temperature, the CO yield increases monotonically for all cases. However, with the addition of additional flue gas to the inlet reaction mixture, the CO yield is reduced. Firstly, this is due to the fact that CO_2 is added in the denominator in Eq. (5) for a fixed amount of methane, while the amount of CO_2 is above the stoichiometric one. Secondly, there is an additional steam in the inlet reaction mixture, which consumes CO for the water-gas shift reaction.

Fig. 8 shows the variation of $\text{H}_2 + \text{CO}$ yield in the product gas. It shows that the formations of $\text{H}_2 + \text{CO}$ increase with increasing methane-to-oxidant ratio of the inlet gas. Moreover, Fig. 8 shows that at the residence time about $4 \text{ kg}_{\text{cat}} \text{ s/mol}$ the addition of flue gas to the inlet mixture have noticeable effect for the moderate low temperature range. For example, If for a temperature of 800°C the discrepancy is about 7%, then for a temperature of 500°C the discrepancy reaches more than 25%.

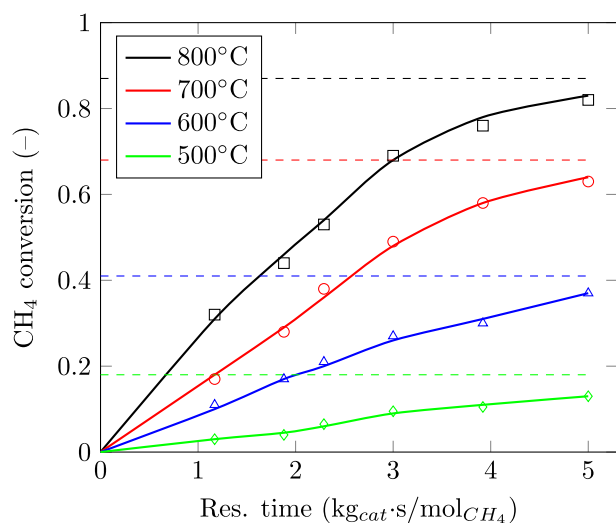
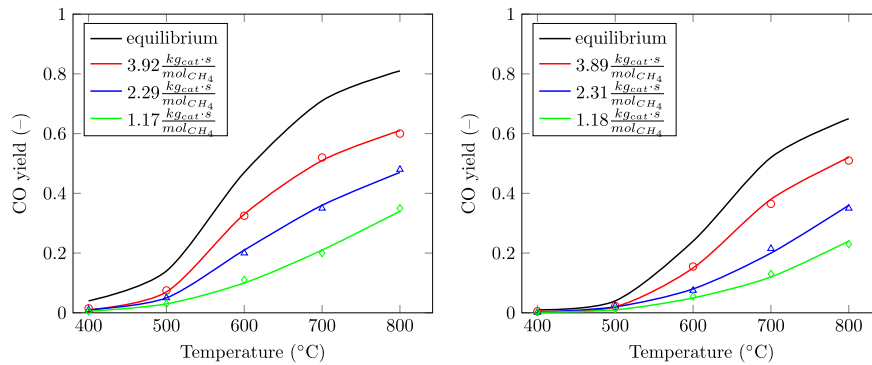
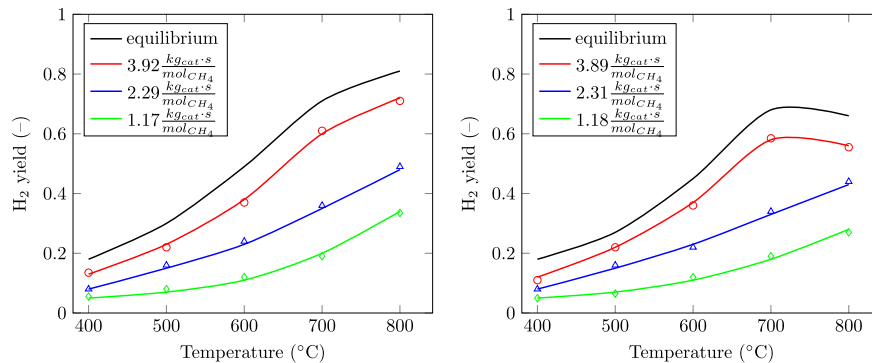


Fig. 5 – Methane conversion as a function of residence time at different temperature, $\text{CH}_4:1/3\beta$, pressure 1 bar. Hollow points – experimental results; solid lines – approximation curves.

Table 2 – CO₂ and H₂O conversion at 3.92 kg_{cat} s/mol for p = 1 bar and CH₄:1/3β

Catalyst shape	CO ₂ conversion				H ₂ O conversion			
	500°C	600°C	700°C	800°C	500°C	600°C	700°C	800°C
Equilibrium	1.8	10.2	56.3	88.6	32.4	56.0	71.1	89.5
7-holes cylinder	1.1	8.6	51.2	84.3	27.1	52.2	67.2	84.9

**Fig. 6 – Effect of temperature on the hydrogen yield at p = 1 bar for the various inlet gas compositions: a) CH₄:1/3β; b) CH₄:2/3β. Hollow points – experimental results; solid lines – approximation curves.****Fig. 7 – Effect of temperature on the carbon monoxide yield at p = 1 bar for the various inlet gas compositions: a) CH₄:1/3β; b) CH₄:2/3β. Hollow points – experimental results; solid lines – approximation curves.**

Effect of pressure

According to the Le Chatelier-Brown principle, the methane reforming process with steam and carbon dioxide is affected not only by the feed gas composition and the temperature but also by pressure. This is due to the fact that both steam and carbon dioxide reforming of methane proceeds with an increase in the volume of the reaction products. Using a leak-proof burner with a tangential supply of methane allows to obtain the methane combustion at various pressures. Therefore, the effect of pressure on the methane reforming process has been studied.

In this study, the methane reforming process has been investigated for the high residence time 3.92 kg_{cat} s/mol at pressure of 1, 3 and 5 bar for the stoichiometric methane-to-oxidant ratio CH₂:1/3β. Fig. 9 shows the effect of pressure on the methane conversions as a function of temperature with CH₂:1/3β. Fig. 9 presents that the methane conversions are suppressed by increasing pressure within the temperature

range under study. With increasing pressure, the methane conversion decreases because the total product mole number is larger than that of the inlet reactants. In addition, as shown by Xu and Froment, the rate of the conversion reaction is inversely proportional to the pressure due to diffusion limitation [48]. Moreover, the same effect of pressure on the methane reforming reaction was confirmed by Hallajbashi and co-authors [49]. However, the effect of pressure is not more than 1–1.8% and in engineering calculations, the effect of pressure at a moderately low pressure (below 5 bar) can be considered insignificant.

Analyzing the results in Fig. 9 it can be argued that the effect of pressure on the yield of hydrogen will also be negligible in the experimentally investigated pressure range.

Effect of steam addition

The experimental setup shown in Fig. 1, allows to investigate the effect of the steam addition on the reforming process. The

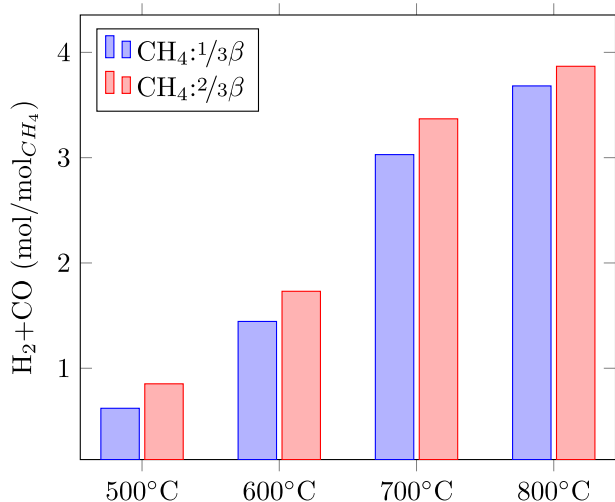


Fig. 8 – Effect of temperature on H₂ + CO molar yield at p = 1 bar for the various inlet gas compositions at 3.92 kg_{cat} s/mol: a) CH₂:β; b) CH₂:2/3β.

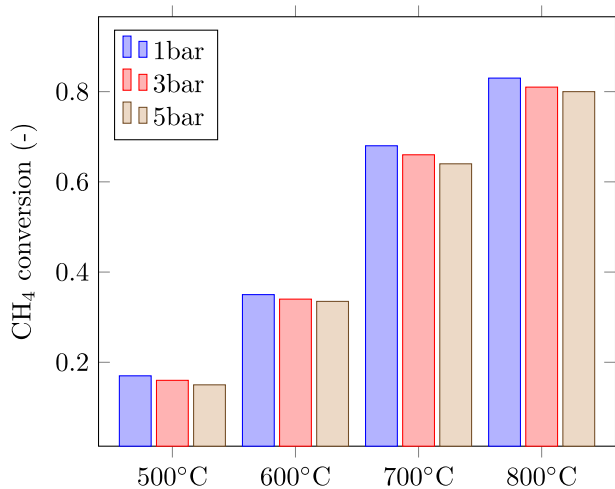


Fig. 9 – Effect of pressure on the methane conversion at CH₂:1/3β for the various temperatures at 3.92 kg_{cat} s/mol.

study of the effect of the steam addition is an important task for the development of thermochemical waste-heat recovery systems that use the combined methane reforming with steam and flue gases [20,21,50].

Fig. 10 shows the changes of the methane conversion as a function of temperature when different steam content is fed into the methane reformer at a fixed CH₂:1/3β ratio at 1 bar when Raschig ring is used as a catalyst. The steam addition has a significant effect on the methane conversion. The steam addition to the inlet reaction gas mixture increases the methane conversion considerably; especially at the temperature range from 500 to 800 °C. This fact is well correlated with Hegarty's [51] and Ozkara-Aydinoglu's [11] articles.

However, a constant increase of the steam amount in the inlet reaction mixture, although it increases the methane conversion, leads to a decrease in the efficiency of the reforming process. First, for the production of steam, additional heat is expended. Secondly, an increasing of the mass

flow of the reaction mixture through the reaction space leads to the appearance of additional aerodynamic resistance. Determination of the optimum amount of steam for the experimental scheme shown in Fig. 1, is an important task for the engineering calculations.

Addition steam has a strong effect on the syngas composition, especially at moderate low temperature. Fig. 11 show the dependence of H₂/CO ratio vs. temperature for a fixed CH₂:1/3β ratio with addition various volume of steam for about 4 kg_{cat} s/mol. The profile changes notably when steam is added to the feed stock. In the temperature range below 600 °C, high hydrogen-to-carbon monoxide ratios are obtained because water-gas shift reaction takes place and steam methane reforming reaction is dominant. With increasing temperature; the H₂/CO ratio is practically effected by both steam and carbon dioxide methane reforming reactions. The increasing of H₂/CO ratio by steam addition is not surprising due to steam methane reforming reaction increases the hydrogen production, and, additionally, the presence of additional steam consumes CO when water-gas shift reaction is performed.

Pressure drop and loss factor

The catalyst particles are placed into the tube with diameter of 100 mm. The catalysts layer depth is 600 mm as presented in Fig. 1. The experiments were performed for three temperatures: 500 °C, 600 °C and 800 °C. The results of experimental investigation of the pressure drops are presented in Fig. 12. In Fig. 12, velocity is calculated via mass flow rate and gas density, which depend on the gas composition and temperature.

In Fig. 12, the experimental data are denoted by the hollow points and the approximating curve of the experimental results are represented by the solid lines. Fig. 12 shows that with increasing temperature the pressure loss decreases. This is due to the fact that the pressure losses inversely depend on the density of the reactants in the methane reformer. In turn, the density of gases depends on temperature. In addition, according to Fig. 4, it can be seen that at

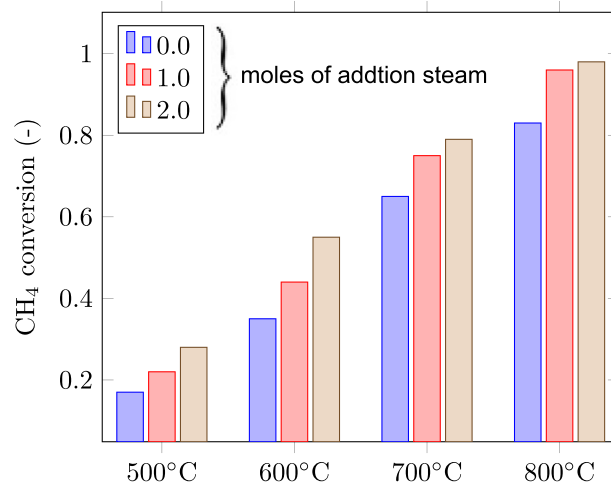


Fig. 10 – Effect of the steam addition on the methane conversion at fixed CH₂:1/3β ratio.

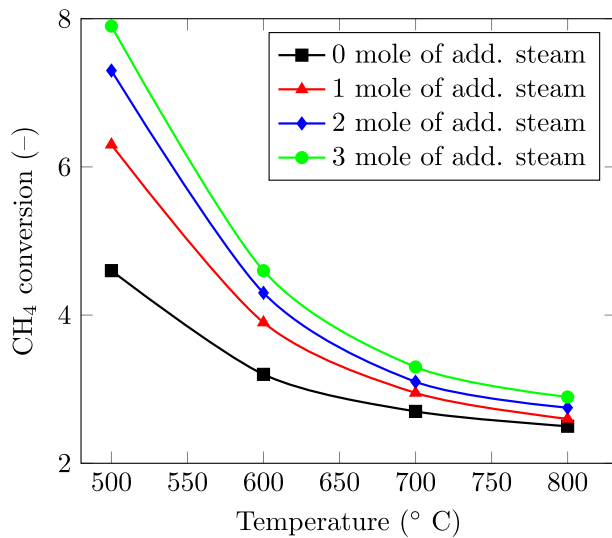


Fig. 11 – Effect of the steam addition on H_2/CO ratio at fixed $CH_4:1/3\beta$ ratio.

moderately low temperatures, the methane conversion rate is low. Therefore, the products of reforming contain mostly unreacted initial elements (methane, steam, carbon dioxide and chemically inert nitrogen). While, as the temperature increases, the mole fraction of hydrogen and carbon monoxide increases. It is known that the density of hydrogen and carbon monoxide is significantly lower than the density of methane, steam and carbon dioxide. For the foregoing reasons, it can be argued reasonably that with increasing reforming temperature the pressure loss decreases. However, it should be considered that this fact is valid only in the case of inertial flow (high-velocities) conditions, for which the Burke-Plummer equation is valid [52]. Eq. (6), which is a simplified representation of that Burke-Plummer equation [53], indicates that pressure drops and gas density respect a direct proportionality relation at a given superficial gas

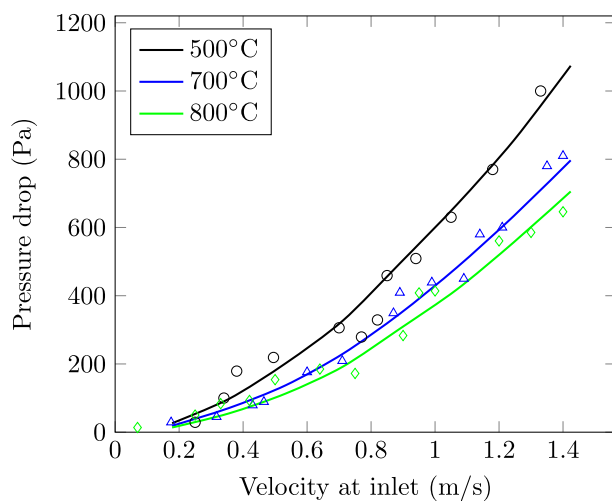


Fig. 12 – The experimental data of pressure drop for various temperatures at $p = 1$ bar and fixed $CH_4:1/3\beta$ ratio: the hollow points – experiment; the solid points and lines – CFD modeling.

velocity, opposite to the inverse dependence stated by the author. In viscous flow (laminar flow) conditions, effects of temperature on fluid viscosity becomes relevant (Blake-Kozeny equation, which constitutes the well-known Ergun equation, summed with Burke-Plummer [53]). Moreover, the variation of molar flow due to the conversion in reaction with overall gaseous moles variation has an effect on the pressure drop. The variation of molar flow due to the SMR and DMR reactions is increasing (at a moderate high temperature range) the overall volumetric flow passing through the packed bed, therefore superficial velocity and consequently measured pressure drops.

From the point of view of the classical representation of fluid dynamics, the pressure losses in the catalyst bed can be considered as the local hydraulic losses, which are caused by changes in the shape and size of the channel, which is deforming the flow. As shown by the experimental results (Fig. 12), it can be approximately assumed that the pressure loss through the catalyst bed is proportional to the square of the velocity. For this reason, for the engineering calculations, it is convenient to characterize the fluid resistance of a catalyst bed with a dimensionless quantity K_l , which is called the local resistance coefficient or loss factor.

The local resistance coefficient can be determined experimentally:

$$K_l = \frac{2\Delta P}{\rho v^2}, \quad (6)$$

where ρ – the density of fluid; v – the velocity of fluid.

Based on the results which are presented in Fig. 12, and also on the basis of the expression (6), the values of the local resistance coefficients were determined. The variations of the local resistance coefficient for 1 m of the packed bed versus the velocity at inlet for the different temperatures are shown in Fig. 13.

As seen in Fig. 13, the loss factor K_l varies substantially when the velocity varies, especially in the range up to 1 m/s. Usually, when flowing around solid non-porous bodies, the coefficient K_l depends weakly on the velocity. This effect of velocity on the coefficient K_l can be explained by the fact that

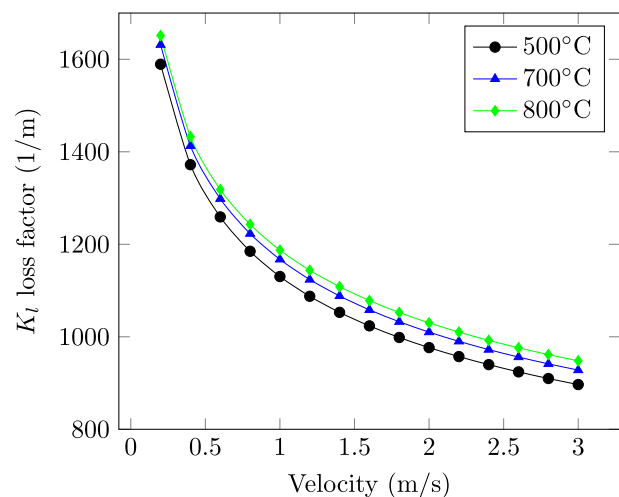


Fig. 13 – The variations of the local resistance coefficients versus the velocity for the different temperatures.

with increasing velocity, the amount of gas that passes through the pores in catalyst increases. At the same time, when the gas velocity is low, the fluid impulse is not enough to percolate through the porous structure. In addition, Fig. 13 shows that the coefficient K_1 is weakly dependent on the reforming temperature. This is explained by the fact that, from the point of view of the classical representation of fluid dynamics, the coefficient K_1 does not depend on the density of the gas in a wide range of Reynolds numbers.

Conclusion

An experimental study of methane reforming with flue gases has been performed on Ni-Al₂O₃ catalyst over wide range of operational parameters such as temperature, methane-to-flue gas ratio, pressure and residence time.

The type of the dependence of the methane conversion, hydrogen and carbon monoxide yield as a function of temperature, residence time at various inlet gas compositions, types of catalysts, and pressures was established experimentally. For the high residence time (about 4 kg_{cat} s/mol), the methane conversion reaches a value that is close to equilibrium. The methane conversion is almost directly proportional to residence time at the temperature range from 600 to 800°C. The effect of pressure on the methane conversion in the pressure range from 1 to 5 bar was investigated. It is established that with increasing pressure, the methane conversion decreases. However, such a reduction is no more than 1–2% for pressures of 1 and 5 bar.

A study of the steam addition to the methane conversion was carried out. The syngas composition can be controlled by addition of steam. When adding steam, the methane conversion increases with fixed operating parameters, while H₂/CO ration decreases.

The pressure drop in the reformer depends on the temperature for the constant velocity of the reaction mixture at the reformer inlet. A characteristic quadratic dependence $\Delta P = f(v^2)$ is observed for all temperatures. the loss factor K_1 is weakly dependent on the reforming temperature, while it is strongly dependent on the gas velocity. The coefficient K_1 decreases with increasing velocity. In the velocity range from 0.5 m/s to 3 m/s, it changes more than twice.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2019.01.250>.

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